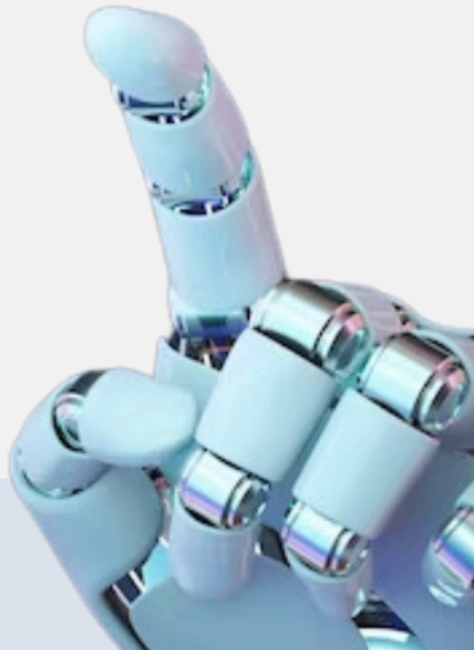
Leveraging Agentic AI Middleware for Intelligent Interaction and Modernization of Legacy Enterprise Applications



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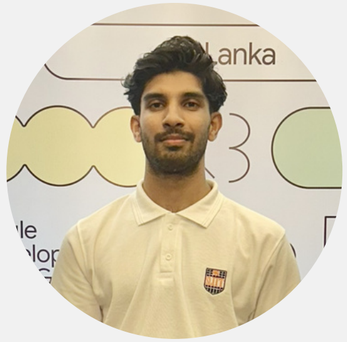

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Problem & Motivation

Context

Many enterprises still rely on legacy systems with outdated interfaces.

Issue

- Complex navigation
- Poor usability
- Difficult to integrate modern technologies

Impact

- Reduced productivity
- High modernization cost
- Slow digital transformation

Research Gap & Objective

Research Gap
Most modernization approaches require rebuilding or replacing entire systems.

Objective
Develop an Agentic AI Middleware that:

- Understands legacy interfaces
- Generates a modern dynamic UI
- Improves usability without modifying the legacy system

Solution Overview

Solution
An AI-powered middleware layer that transforms legacy interfaces into modern dynamic user interfaces.

Key Idea
Modern UI → AI Middleware → Legacy System

Result
Better usability without rebuilding the system.

Key Features

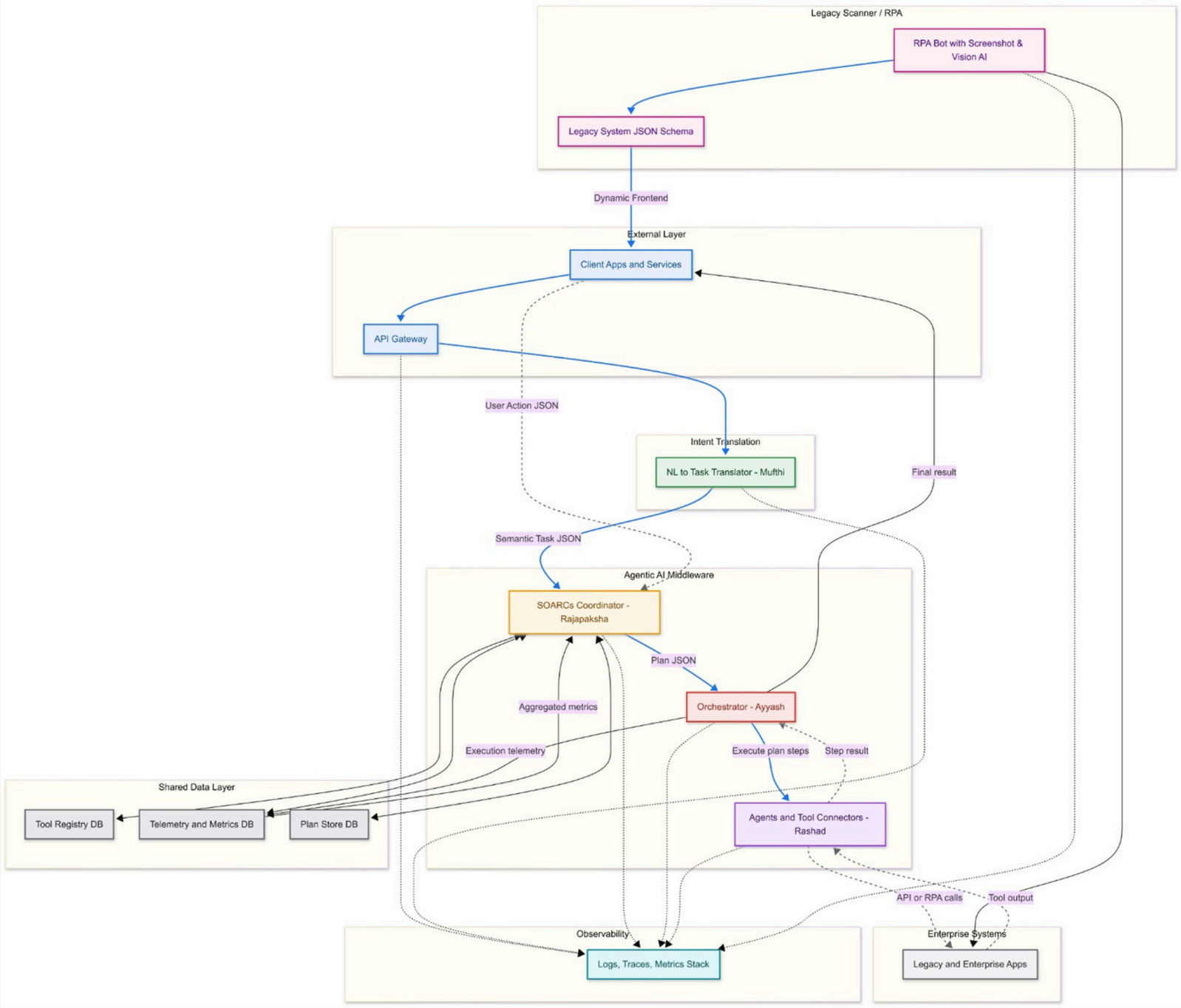
Dynamic UI Generation
Automatically creates modern interfaces.

Workflow Simplification
Reduces complex legacy operations.

Middleware Integration
Connects new UI with legacy system functions.

Non-Intrusive Modernization
Legacy system remains unchanged.

System Diagram



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Technology Stack

Frontend
React vite, Tailwind CSS, Electron

Database & Storage
MongoDB , Cloudinary

Backend
Node.js, Python FastAPI

Integration
WebHook, Web Sockets

AI Layer
Lama LLM

Commercialization

Target Users
Organizations using legacy systems (Banks, Government, Healthcare)

Value
- Modern UI without system replacement
- Reduced modernization cost
- Improved employee productivity

Business Model
Enterprise software licensing / SaaS.

Thank You !

Any Questions ?